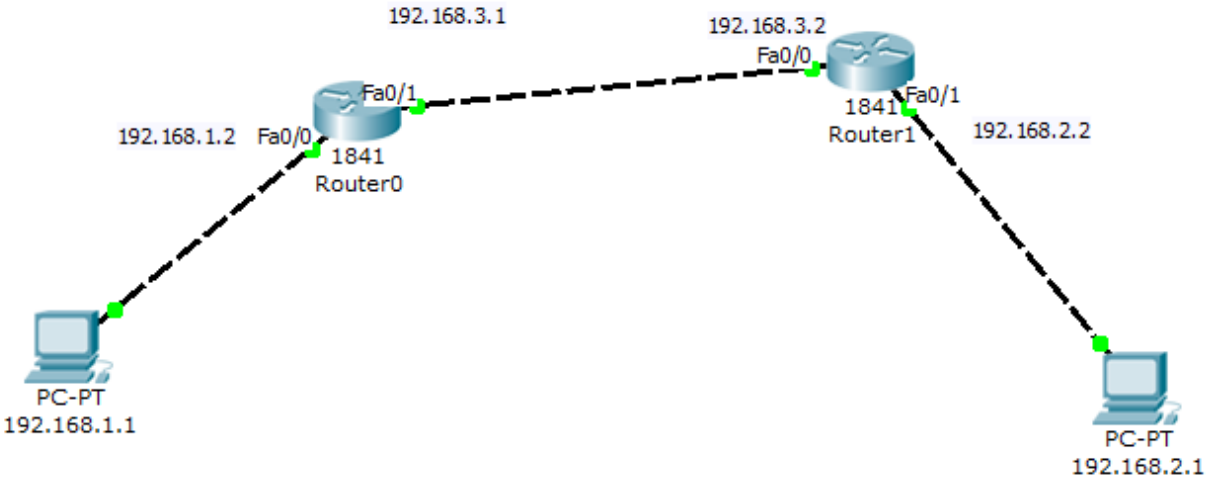
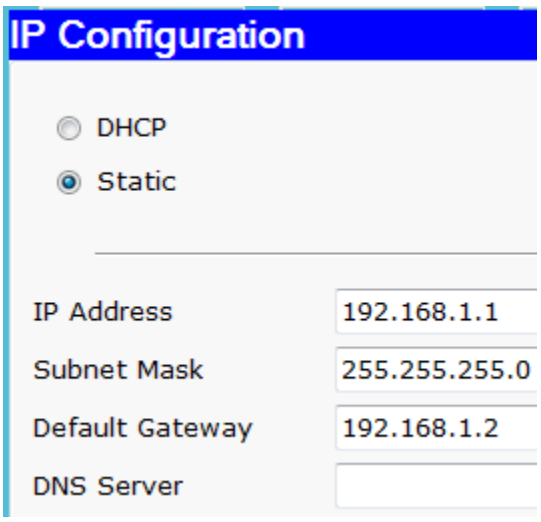
Expt-4
Connection of Two Routers

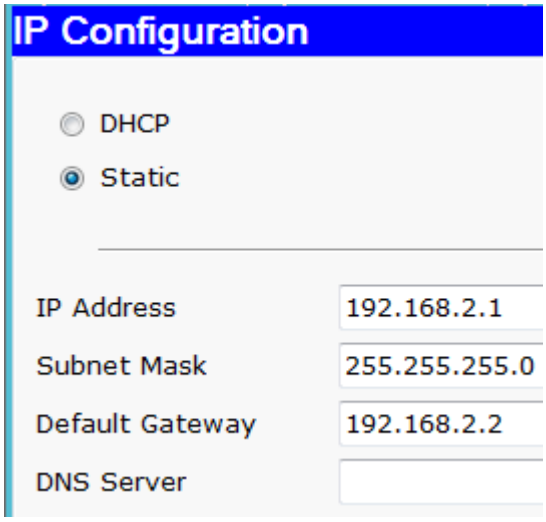
This experiment can be considered as the extended version of experiment 4. Here a packet will be routed through two routers and configuration of two or more routers in series is very complicated than the case of a single router.

Step-1

Implement the network shown below where the IP address and the corresponding interfaces are also shown for convenience.







Step-1

Let us configure the router 0

```
Continue with configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#ip add 192.168.1.2 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int fa 0/1
Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr
Building configuration...
[OK]
Router#
```

Step-2

Configure the Router-1

```
Continue with configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa 0/0
Router(config-if)#ip add 192.168.1.2 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#exit
Router(config)#int fa 0/1
Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config-if)#no shut
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router#wr
Building configuration...
[OK]
Router#
```

Step-3
Configure the Router-0 again

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.2.0 255.255.255.0 192.168.3.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
Gateway of last resort is not set
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 192.168.3.2
C    192.168.3.0/24 is directly connected, FastEthernet0/1
Router#
```

Step-4
Configure the Router-1 again

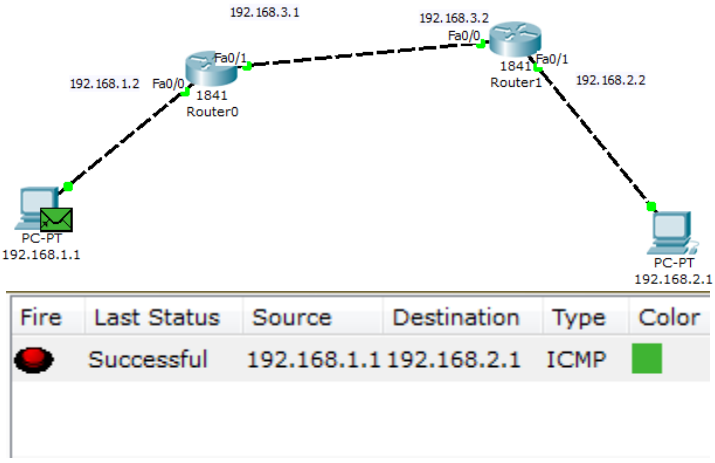
```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 192.168.3.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
S 192.168.1.0/24 [1/0] via 192.168.3.1
C 192.168.2.0/24 is directly connected, FastEthernet0/1
C 192.168.3.0/24 is directly connected, FastEthernet0/0
Router#

Step-6
Test the Router-1 like:

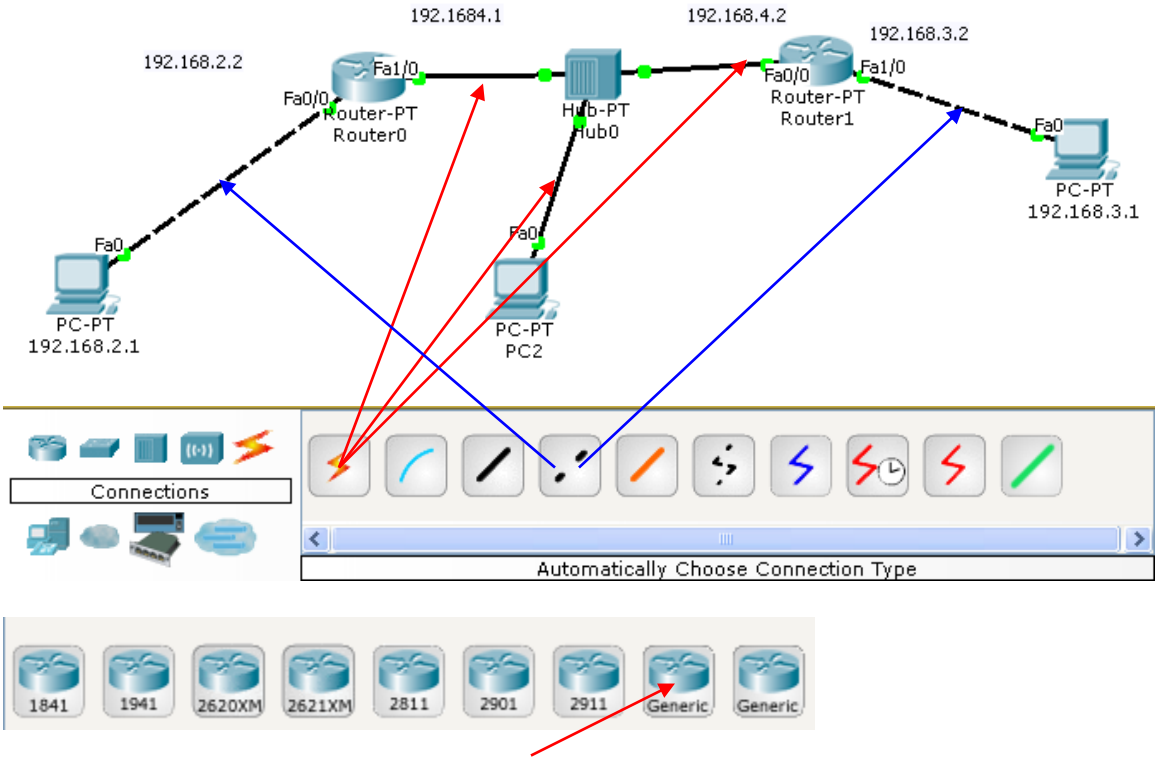
Router>en
Router#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 62/62/63 ms
Router#

Step-7
Verify the ICMP packet routing from simulation mode.



Connection two routers and incorporation of host between them

1. Prepare the following network.



2. Configure IP of each of the PC.

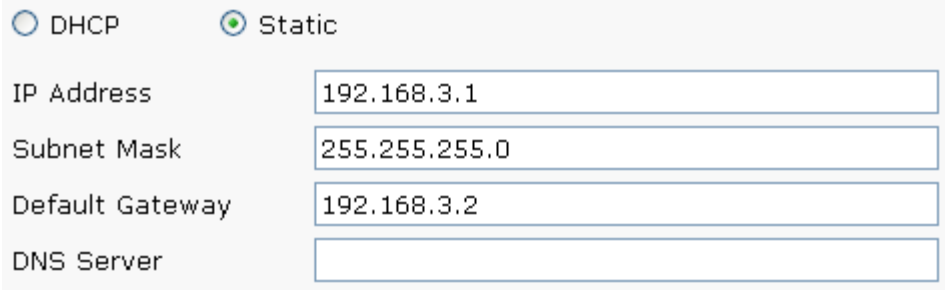
IP Configuration

DHCP Static

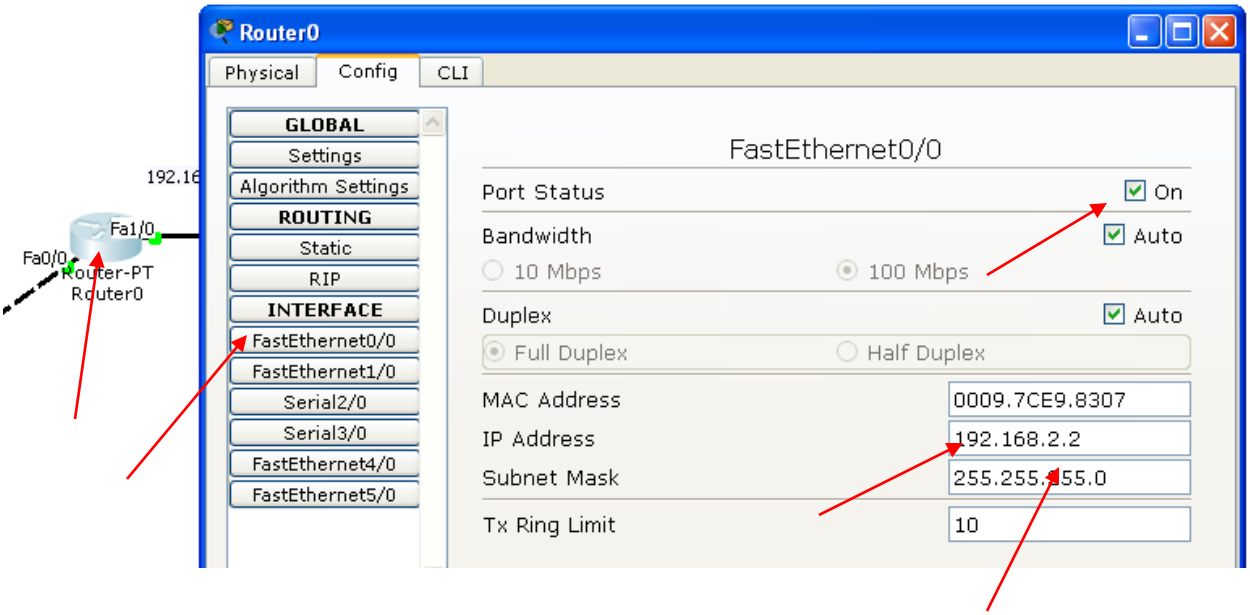
IP Address	192.168.2.1
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.2
DNS Server	

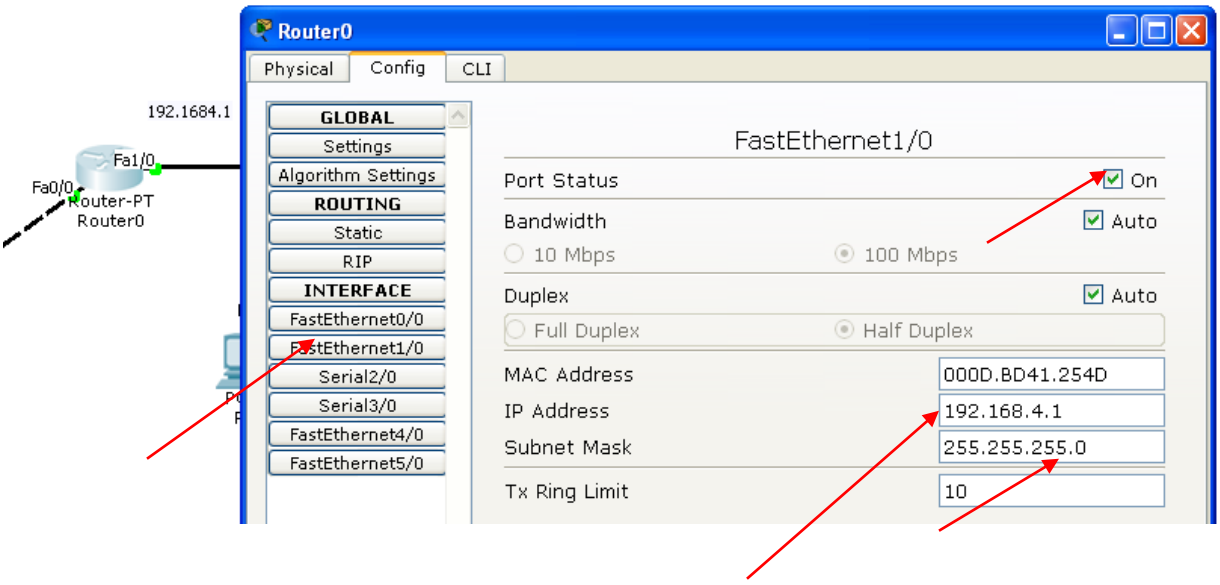
DHCP Static

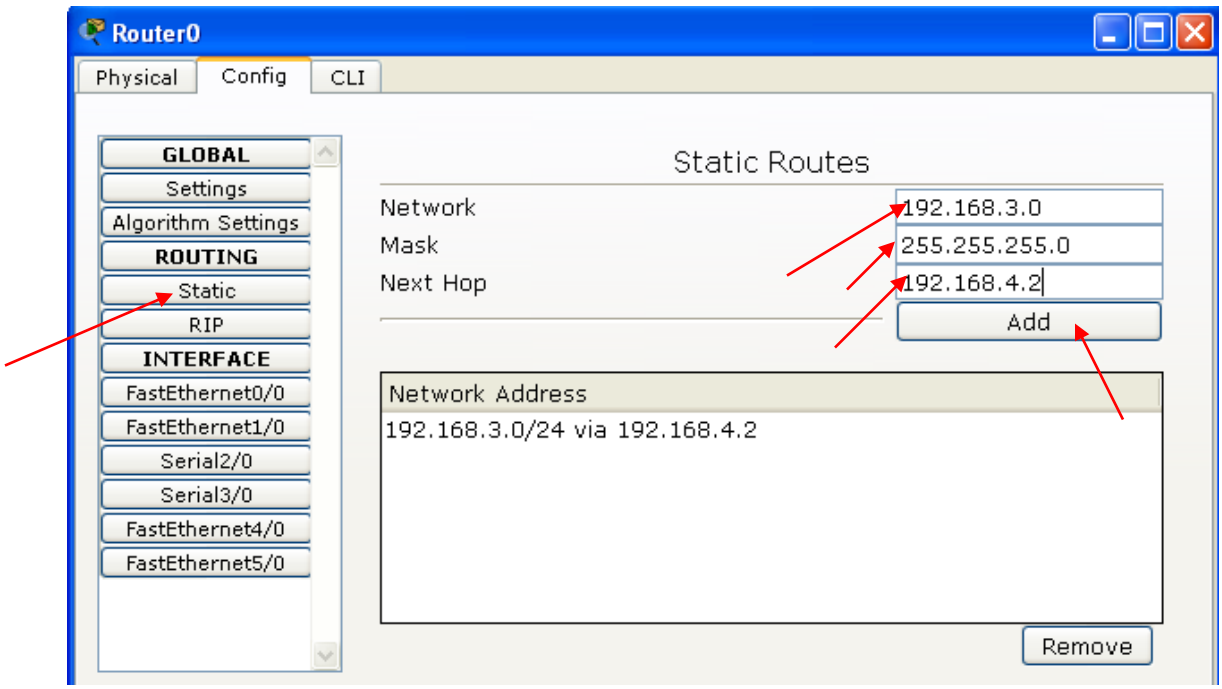
IP Address	192.168.4.4
Subnet Mask	255.255.255.0
Default Gateway	192.168.4.1
DNS Server	



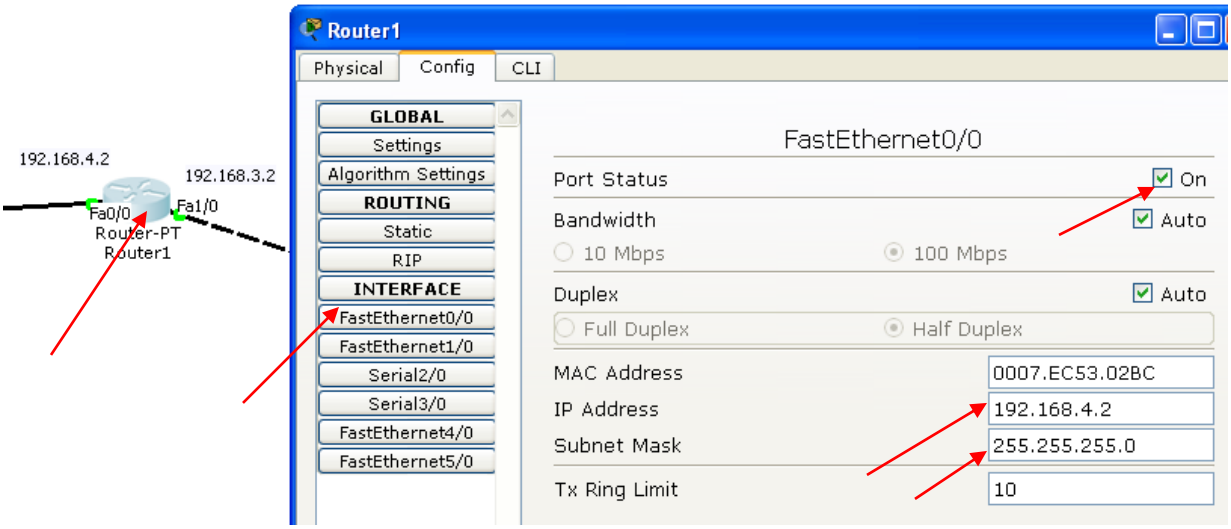
3. Configure the Router-0

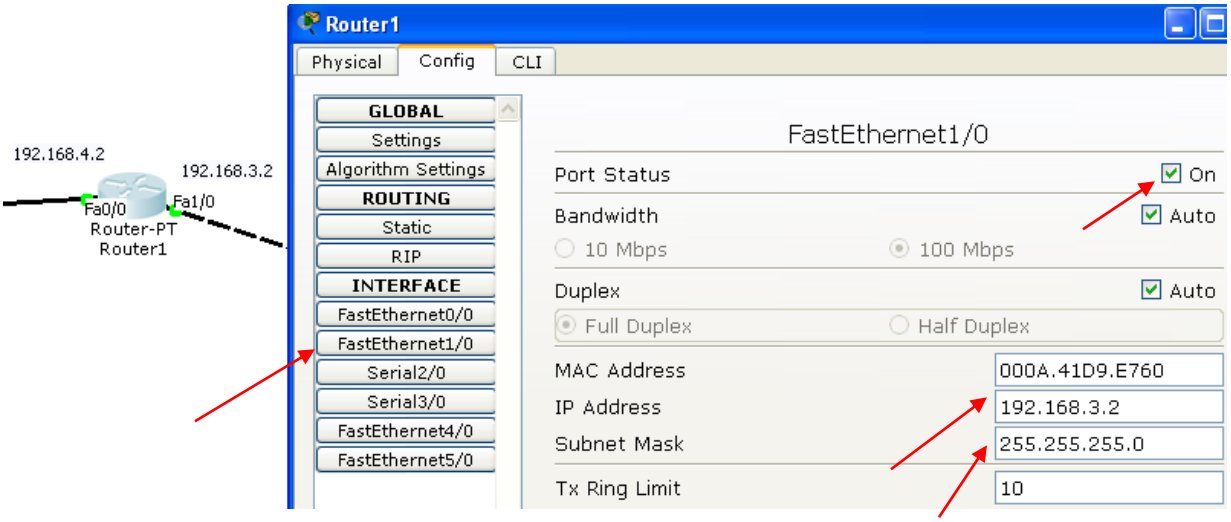


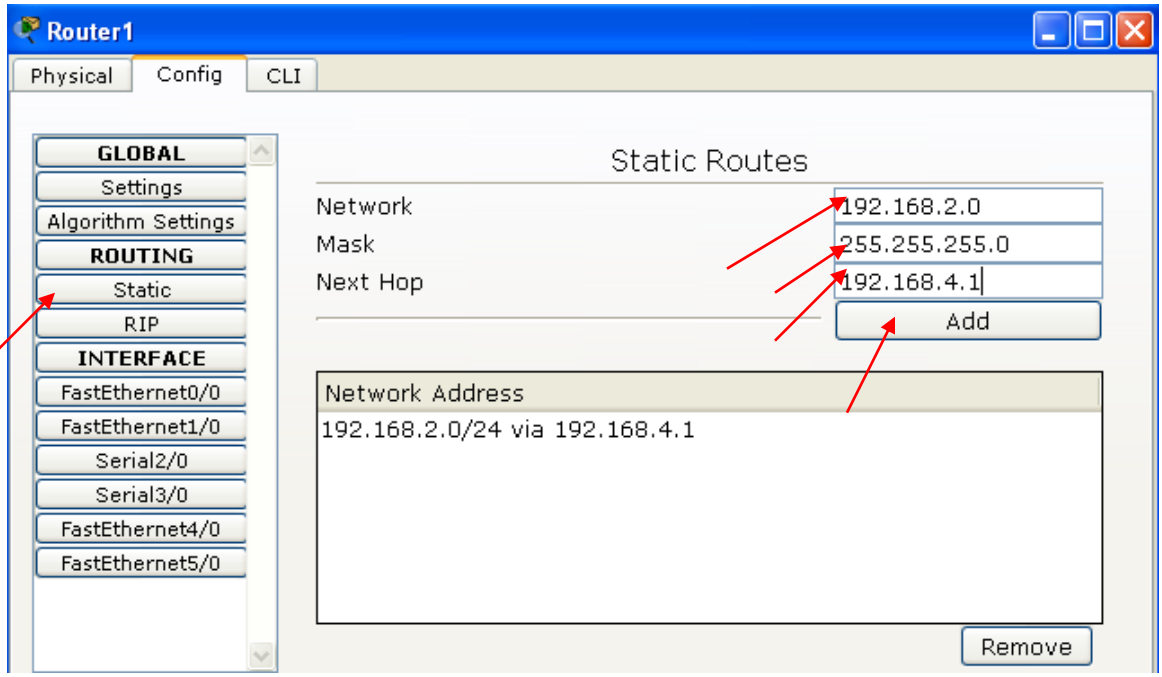




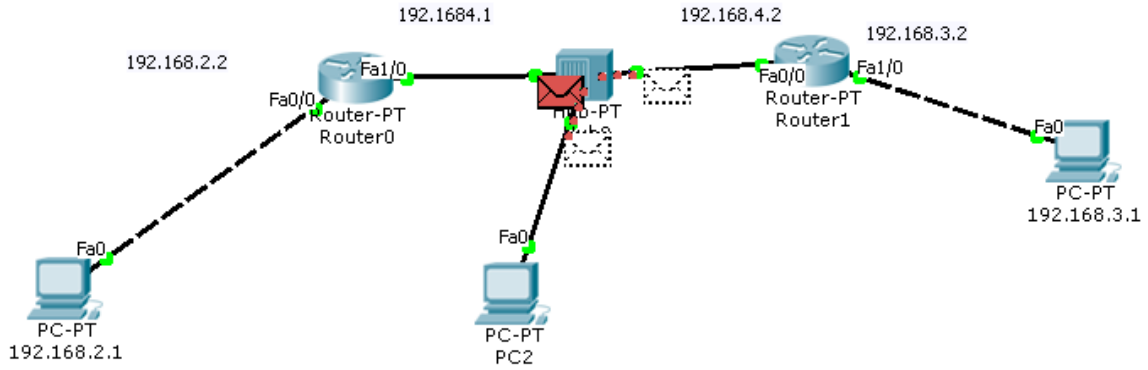
4. Configure the Router1



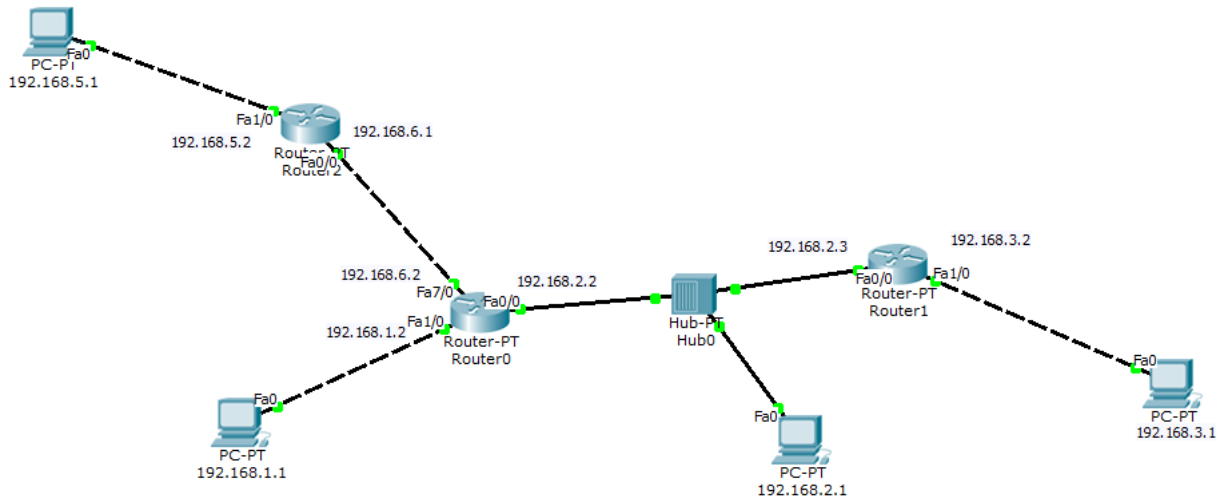


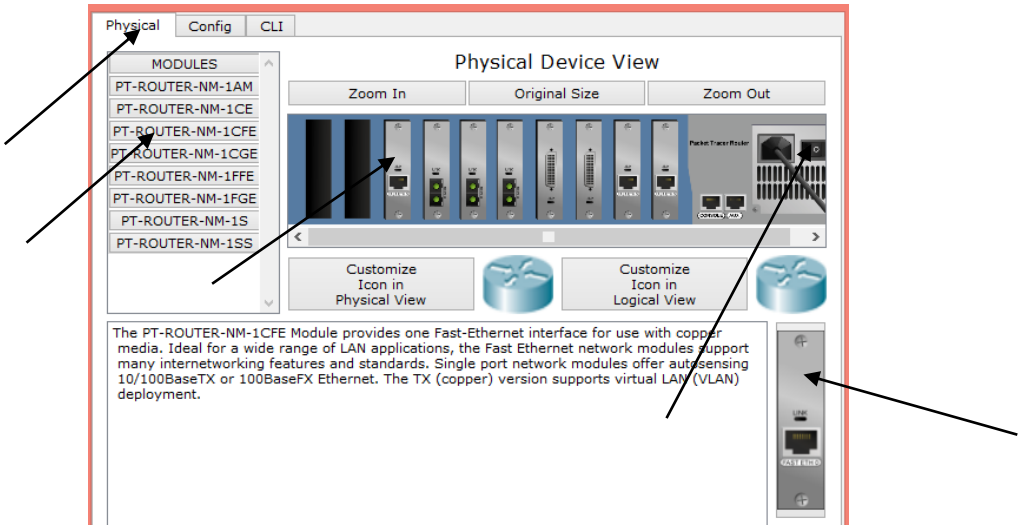


5. Verify the network.

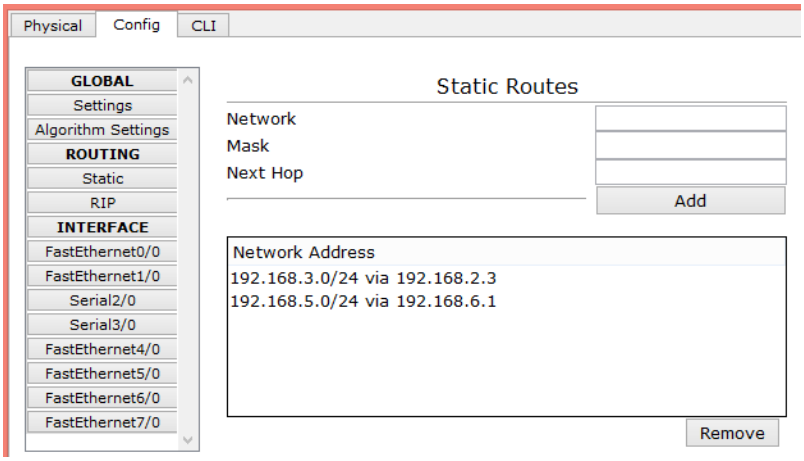


Example-1

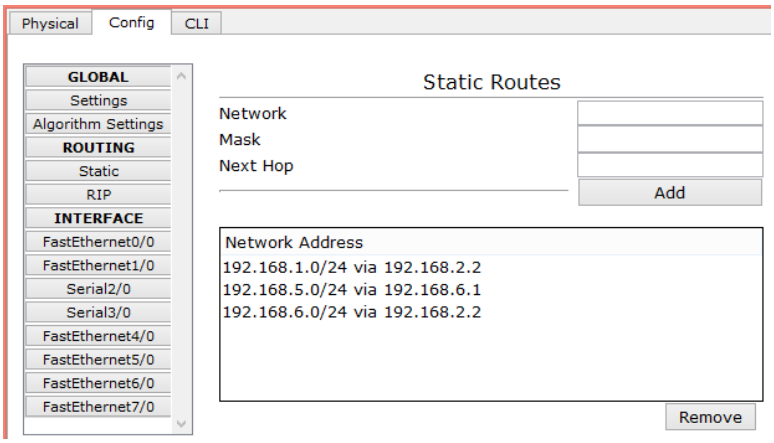




Router 0



Router 1



Router 2

